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# क्षणिक Kshanik

"क्षणिक" — transient, momentary · Physics-Informed Yield Surrogate for a Non-Isothermal Plug-Flow Reactor

A machine learning surrogate for reactor yield prediction, built by fitting the governing reaction kinetics and energy balance directly, then applying **Gaussian Process** and **CatBoost** models to correct only the residual error the mechanistic fit leaves behind.

Cross-validated RMSE 8.79 on 150 training observations

Team Transience · Fugacity 2026 ML Hackathon, IIT Kharagpur

· Vaishnav Aditya · Preenithi Varshana · Kritika Pandey · 150 training rows, 50 held-out test rows

train\_dataset.csv → mechanistic\_pfr.py → warped GPR / CatBoost → overall\_yield

[GitHub Repository ↗](#)[Full Technical Report \(PDF\) ↗](#)

## 25%

ROWS AT EXACT ZERO  
YIELD

## 5 → 13

RAW INPUTS → PHYSICS  
FEATURES

## 8.79

BEST CV RMSE (WARPED  
GPR)

## 2

MODELS: SMALL-DATA  
& LARGE-DATA

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## LIMITATION OF DIRECT REGRESSION

## Black-Box Models Cannot Isolate the Yield Collapse

A regressor fit directly on `flow_rate`, `concentration`, `inlet_T`, `length`, and `jacket_T` must infer, from 150 rows alone, that yield falls to exactly **0%** whenever residence time and temperature drive the side reaction ( $B \rightarrow C$ ) past the desired one ( $A \rightarrow B$ ). No single raw feature correlates strongly with yield — the highest is  $|r| = 0.64$  — because the governing variable is a non-linear combination the model has no basis to construct from data this limited.

## TRANSIENCE APPROACH

## A Mechanistic Prior, Corrected by a Gaussian Process

We fit a non-isothermal PFR model directly — series kinetics  $A \rightarrow B \rightarrow C$  under Arrhenius rate laws, coupled to a jacket energy balance — using nonlinear least squares on the training set. Its output serves as the **prior mean** for a Gaussian Process, which is left to learn only the mechanistic model's residual error rather than the full yield surface.

MODEL PERFORMANCE —  $5 \times 10$  REPEATED K-FOLD CVMECHANISTIC  
MODEL ALONE**12.17**RMSE · physics  
only, no MLADDITIVE-  
RESIDUAL GPR**10.14**RMSE · mechanistic  
mean + plain GPWARPED-LOGIT  
GPR**8.79**RMSE · selected  
small-data modelCATBOOST +  
SYNTHETIC**11.35**RMSE · large-data  
model

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THE GOVERNING SYSTEM — FIT DIRECTLY TO TRAINING DATA

|                                                                                                           |                                                                                                                 |                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MOLE BALANCES</b><br>$\frac{dC_A}{d\tau} = -k_1(T) C_A$ $\frac{dC_B}{d\tau} = k_1(T) C_A - k_2(T) C_B$ | <b>ARRHENIUS KINETICS</b><br>$k_i(T) = k_{0,i} \exp(-E_{a,i} / RT)$ $i \in \{1, 2\} \text{ — one per reaction}$ | <b>ENERGY BALANCE</b><br>$\frac{dT}{d\tau} = [U_a(T_j - T) + (-\Delta H_1)r_1 + (-\Delta H_2)r_2] / \rho C_p$ <p><i>jacket heat exchange + reaction enthalpy of both steps</i></p> |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Eight effective parameters —  $k_{0,1}$ ,  $E_{a,1}$ ,  $k_{0,2}$ ,  $E_{a,2}$ ,  $U_a$ ,  $\Delta H_1$ ,  $\Delta H_2$ ,  $\rho C_p$  — are recovered from the 150 training rows via multi-start nonlinear least squares, not assumed from literature values. The fitted result:  $E_{a,2} > E_{a,1}$ , meaning the side reaction accelerates with temperature faster than the desired one — the direct physical cause of the zero-yield cliff seen in the training data.

#### MECHANISTIC FOUNDATION

### Non-Isothermal PFR Model

Series reaction  $A \rightarrow B \rightarrow C$  under Arrhenius kinetics  $k_1(T)$ ,  $k_2(T)$ , coupled to a jacket energy balance with explicit reaction-heat terms (Fogler, *Elements of Chemical Reaction Engineering*, Ch. 8). Fit as a grey-box model via multi-start nonlinear least squares, not assumed from literature constants.

Train RMSE: **12.17**

- $E_{a_2} > E_{a_1}$ , consistent with side-reaction dominance at high temperature

#### SMALL-DATA MODEL

### Warped Gaussian Process

The GP is trained in  $\text{logit}(\text{yield}/100)$  space rather than on the raw residual. This is the statistically appropriate likelihood for a target bounded on  $[0, 100]$  with substantial mass at the boundary, consistent with the censored-regression and warped-GP literature.

CV RMSE: **8.79**  $\pm$  3.87

- outperforms both additive-residual and classifier-gated variants

#### LARGE-DATA MODEL

### CatBoost with Synthetic Augmentation

5,000 synthetic observations, sampled across the operating envelope and simulated through the fitted ODE, are combined with the real data at a  $0.15\times$  training weight. A classifier stage first separates the collapsed-yield regime before a regressor is applied, a strategy intended to scale to a larger production dataset.

CV RMSE: **11.35** • versus 15.24 on real data alone, no augmentation